Annexure-I

List of Tasks

Task No.	Task Title	Description	Week
1.	Installation	Download and install Anaconda3 Install PyTorch Install TensorFlow 2.0 Install VSCode Install PyCharm	1
2.	Linux Commands	Practice these commands: pwd, cd, ls, cat, sudo, man, redirection, mkdir, rm, rmdir, cp, mv, file, reading, cat, more, less, head, alias, shutdown, restart, touch, nano, bash, sh, chmod, ps, kill, dpkg	1

3.	Python	```python
		# This program adds two numbers
		
		num1 = 1.5
		num2 = 6.3
		
		# Add two numbers
		sum = num1 + num2
		
		# Display the sum
		print('The sum of {0} and {1} is {2}'.format(num1, num2, sum))
		``
		1	

4.	Python	
		`# Store input numbers`
		`num1 = input('Enter first number: ')`
		`num2 = input('Enter second number: ')`
		
		`# Add two numbers`
		`sum = float(num1) + float(num2)`
		
		`# Display the sum`
		`print('The sum of {0} and {1} is {2}'.format(num1, num2, sum))`
			1

5.	Python	
		`# Python Program to calculate the square root`
		
		`# Note: change this value for a different result`
		`num = 8`
		
		`# To take the input from the user`
		`#num = float(input('Enter a number: '))`
		
		`num_sqrt = num ** 0.5`
		`print('The square root of %0.3f is %0.3f'%(num ,num_sqrt))`
			1

6.	Python	
		`# Find square root of real or complex numbers`
		`# Importing the complex math module`
		`import cmath`
		
		`num = 1+2j`
		
		`# To take input from the user`
		`#num = eval(input('Enter a number: '))`
		
		`num_sqrt = cmath.sqrt(num)`
		`print('The square root of {0} is {1:0.3f}+{2:0.3f}j'.format(num ,num_sqrt.real,num_sqrt.imag))`
			1

7.	Python	
		`# Python Program to convert temperature in celsius to fahrenheit`
		
		`# change this value for a different result`
		`celsius = 37.5`
		
		`# calculate fahrenheit`
		`fahrenheit = (celsius * 1.8) + 32`
		`print('%0.1f degree Celsius is equal to %0.1f degree Fahrenheit' %(celsius,fahrenheit))`
			1

8.	Python		1

```python
# Python Program to find the area of triangle

a = 5
b = 6
c = 7

# Uncomment below to take inputs from the user
# a = float(input('Enter first side: '))
# b = float(input('Enter second side: '))
# c = float(input('Enter third side: '))

# calculate the semi-perimeter
s = (a + b + c) / 2

# calculate the area
area = (s*(s-a)*(s-b)*(s-c)) ** 0.5
print('The area of the triangle is %0.2f' %area)
```

9.	Python	
		`# Solve the quadratic equation ax**2 + bx + c = 0`
		
		`# import complex math module`
		`import cmath`
		
		`a = 1`
		`b = 5`
		`c = 6`
		
		`# calculate the discriminant`
		`d = (b**2) - (4*a*c)`
		
		`# find two solutions`
		`sol1 = (-b-cmath.sqrt(d))/(2*a)`
		`sol2 = (-b+cmath.sqrt(d))/(2*a)`
		
		`print('The solution are {0} and {1}'.format(sol1,sol2))`
			1

10.	Python	
		`# Taking kilometers input from the user`
		`kilometers = float(input("Enter value in kilometers: "))`
		
		`# conversion factor`
		`conv_fac = 0.621371`
		
		`# calculate miles`
		`miles = kilometers * conv_fac`
			1

11.	Python	`i = 10`
		`if (i > 15):`
		`  print ("10 is less than 15")`
		`print ("I am Not in if")`
			1

12.	Python	<pre>i = 20;
		if (i < 15):
		    print ("i is smaller than 15")
		    print ("i'm in if Block")
		else:
		    print ("i is greater than 15")
		    print ("i'm in else Block")
		print ("i'm not in if and not in else Block")</pre>
			1

13.	Python	<pre>i = 10
		if (i == 10):
		    # First if statement
		    if (i < 15):
		        print ("i is smaller than 15")
		    # Nested - if statement
		    # Will only be executed if statement above
		    # it is true
		    if (i < 12):
		        print ("i is smaller than 12 too")
		    else:
		        print ("i is greater than 15")</pre>
			1

14.	Python	<pre>i = 20
		if (i == 10):
		    print ("i is 10")
		elif (i == 15):
		    print ("i is 15")
		elif (i == 20):
		    print ("i is 20")
		else:
		    print ("i is not present")</pre>
			1

15.	Python	Exercise on for loops in Python: https://www.geeksforgeeks.org/python-for-loops/	1
16.	Python	Exercise on While loops in Python: https://www.geeksforgeeks.org/python-while-loops/	1
17.	Python	Exercise on Break statement in Python: https://www.geeksforgeeks.org/python-break-statement/	1
18.	Python	Exercise on Continue statement in Python: https://www.geeksforgeeks.org/python-continue-statement/	1
19.	Python	Exercise on various looping techniques in Python: https://www.geeksforgeeks.org/looping-techniques-python/	1
20.	Python	Exercise on User defined functions in Python: https://www.geeksforgeeks.org/functions-in-python/	2

21.	Python	Exercise on List data type in Python: https://www.programiz.com/python-programming/list	1
22.	Python	Exercise on Tuple data type in Python: https://www.programiz.com/python-programming/tuple	1
23.	Python	Exercise on String data type in Python: https://www.programiz.com/python-programming/string	1
24.	Python	Exercise on Set data type in Python: https://www.programiz.com/python-programming/set	1
25.	Python	Exercise on Dictionary data type in Python: https://www.programiz.com/python-programming/dictionary	1
26.	Python	Exercise on Exception Handling in Python: https://www.programiz.com/python-programming/exception-handling	2
27.	Python	Exercise on User defined Exception Handling in Python: https://www.programiz.com/python-programming/user-defined-exception	2
28.	Numpy	Exercise on Numpy create Array Using Python: https://www.w3schools.com/python/numpy_creating_arrays.asp	3,4
29.	Numpy	Exercise on Numpy Indexing in Array Using Python: https://www.w3schools.com/python/numpy_array_indexing.asp	3,4
30.	Numpy	Exercise on Numpy Slicing in Array Using Python: https://www.w3schools.com/python/numpy_array_slicing.asp	3,4
31.	Numpy	Exercise on Numpy Slicing in Array Using Python: https://www.w3schools.com/python/numpy_data_types.asp	3,4
32.	Numpy	Exercise on Numpy Array coping and viewing : https://www.w3schools.com/python/numpy_copy_vs_view.asp	3,4
33.	Numpy	Exercise on Numpy Array Shaping : https://www.w3schools.com/python/numpy_array_shape.asp	3,4
34.	Numpy	Exercise on Numpy Array reshaping : https://www.w3schools.com/python/numpy_array_reshape.asp	3,4
35.	Numpy	Exercise on Numpy Array iteration: https://www.w3schools.com/python/numpy_array_iterating.asp	3,4
36.	Numpy	Exercise on Numpy Matrix joining https://www.w3schools.com/python/numpy_array_join Week 4.asp	3,4
37.	Numpy	Exercise on Numpy Array splitting https://www.w3schools.com/python/numpy_array_split.asp	3,4
38.	Numpy	Exercise on Numpy Array searching https://www.w3schools.com/python/numpy_array_search.asp	3,4
39.	Numpy	Exercise on Numpy Array sorting https://www.w3schools.com/python/numpy_array_sort.asp	3,4
40.	Numpy	Exercise on Numpy Array Random technique https://www.w3schools.com/python/numpy_random.asp	3,4
41.	Pandas	Exercise on Pandas basics: https://www.w3schools.com/python/pandas_tutorial.asp	3,4
42.	Pandas	Exercise on Pandas installation: https://www.w3schools.com/python/pandas_getting_started.asp	3,4
43.	Pandas	Exercise on Pandas Series data https://www.w3schools.com/python/pandas_series.asp	3,4
44.	Pandas	Exercise on Pandas Data Frame: https://www.w3schools.com/python/pandas_dataframes.asp	3,4
45.	Pandas	Exercise on Pandas Open CSV files: https://www.w3schools.com/python/pandas_csv.asp	3,4

46.	Pandas	Exercise on Pandas Data analyzation: https://www.w3schools.com/python/pandas_analyzing.asp
47.	Pandas	Exercise on Pandas Data Cleaning techniques: https://www.w3schools.com/python/pandas_cleaning.asp
48.	Pandas	Exercise on Pandas Data Correlation: https://www.w3schools.com/python/pandas_correlations.asp
49.	Stats	Perform Mean, Midian and mode: https://www.w3schools.com/python/python_ml_mean_median_mode.asp
50.	Stats	Perform Standard Deviation: https://www.w3schools.com/python/python_ml_standard_deviation.asp
51.	Machine Learning	Implement Linear Regression https://stackabuse.com/linear-regression-in-python-with-scikit-learn/
52.	Machine Learning	Perform Logistics Regression: https://towardsdatascience.com/logistic-regression-using-python-sklearn-numpy-mnist-handwriting-recognition-matplotlib-a6b31e2b166a https://www.datacamp.com/community/tutorials/understanding-logistic-regression-python
53.	Machine Learning	Exercise on Decision Tree: https://www.datacamp.com/community/tutorials/decision-tree-classification-python
54.	Machine Learning	Exercise on SVM: https://stackabuse.com/implementing-svm-and-kernel-svm-with-pythons-scikit-learn/
55.	Machine Learning	Exercise on Time Series Analysis: https://www.dataquest.io/blog/tutorial-time-series-analysis-with-pandas
56.	Machine Learning	Demonstration of Neural Networks: https://www.analyticsvidhya.com/blog/2020/07/neural-networks-from-scratch-in-python-and-r
57.	Deep Learning	Exercise on MLP: https://machinelearningmastery.com/neural-networks-crash-course/
58.	Deep Learning	Exercise on Feed Forward neural networks: https://builtin.com/data-science/feedforward-neural-network-intro
59.	Deep Learning	Exercise on Neural Network: https://www.analyticsvidhya.com/blog/2019/08/detailed-guide-7-loss-functions-machine-learning-python-code/
60.	Deep Learning	Exercise on Linguistics using Machine learning in python: https://medium.com/towards-artificial-intelligence/natural-language-processing-nlp-with-python-tutorial-for-beginners-1f54e610a1a0
61.	Deep Learning	Text processing: https://pythonspot.com/category/nltk/
62.	Deep Learning	Text Analysis https://medium.com/towards-artificial-intelligence/natural-language-processing-nlp-with-python-tutorial-for-beginners-1f54e610a1a0
63.	Deep Learning	Demonstrate Convolution Neural Network: https://towardsdatascience.com/a-comprehensive-guide-to-convolutional-neural-networks-the-eli5-way-3bd2b1164a53
64.	Deep Learning	Perform CNN on CIFAR-10 Dataset (https://www.analyticsvidhya.com/blog/2020/02/learn-image-classification-cnn-convolutional-neural-networks-3-datasets/)
65.	Microsoft Azure	Microsoft Azure Video Lectures at Microsoft Learning